Gilles FEVRY

PhD candidate

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Education

ENS Paris-Saclay, Master MVA (Mathematics, Vision, Learning) **Sept 2025 – May 2026**

- **Courses:** Optimal Transport, Deep Learning, Optimization & Algorithms, Online Learning.
- **Project:** Labeled Gromov-Wasserstein Optimal Transport for genetic perturbation data.

ENSAE Paris, Engineering cycle, Data Science, Statistics and Learning track **Sept 2023 – May 2026**

- **Rank:** 3^{rd} out of 140. **Courses:** Statistics, Advanced Machine Learning, Reinforcement Learning, Generative Models, Bayesian Statistics, Stochastic Processes, Time Series, Econometrics, Microeconomics, Macroeconomics.
- **Project:** Development of SurvTreeShap(t), an interpretability tool for random forests in survival analysis.

Université de Versailles Saint-Quentin-en-Yvelines, Bachelor's in Mathematics **Sept 2021 – June 2023**

- **Rank:** 2^{nd} out of 60. **Courses:** Measure Theory, Probability, Differential Calculus, Optimization.
- **Project:** Proof of Cramér's theorem and introduction to large deviations.

Professional and Research Experience

PhD candidate, CERMICS – Paris, France **May 2026 – Now**

- PhD in Generative Modeling and Optimisation under the supervision of Pierre-Cyril Aubin and Axel Parmentier.
- Leveraging generative models to sample the set of good solutions to contextual combinatorial optimization problems, with tools spanning Diffusion Models, Flow Matching, Stochastic Localization, and MILP solvers.

Research Intern, Swiss Tropical and Public Health Institute – Basel, Switzerland **June 2025 – Aug 2025**

- Developed an LSTM architecture and framework to emulate OpenMalaria and compared its performance against multivariate Gaussian Process regression.
- Divided simulation computation time by 10^5, enabling large-scale studies, and dashboard deployment.

Statistician Intern, Direction Générale de l'Aviation Civile (DGAC) – Paris, France **June 2024 – Aug 2024**

- Implemented new methods to analyze and visualize the evolution of the Air Passenger Transport Price Index.
- Designed a rotating sampling plan to improve the robustness of the IPTAP.

Publications

SurvTreeSHAP(t): scalable explanation method for tree-based survival models **Aug 2025**

Under the supervision of Lucas Ducrot and Sandrine Katsahian (HeKA, INRIA).

Workshop on Explainable Artificial Intelligence (XAI), IJCAI 2025. https://hal.science/hal-05108033v1

Skills

Programming: Python (Pandas, NumPy, Scikit-Learn, PyTorch, TensorFlow, Selenium, Gradio), R, SAS.

Languages: English C2 (6 years in South Africa), Portuguese C1 (12 years in Portugal), German B1.

Interests

Sectors: Healthcare, Aviation.

Sports: Scuba diving (Advanced), Badminton, Rock climbing.